CHENIERE

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Submitted via www.regulations.gov

Re: **Docket ID No. PHMSA–2019–0091 -** Response to PHMSA's Advanced Notice of Proposed Rulemaking on Pipeline Safety: Amendments to Liquefied Natural Gas Facilities

Dear Acting Administrator Benjamin D. Kochman,

Cheniere Energy, Inc. (Cheniere) appreciates the opportunity to submit the following comments in response to the Pipeline Safety: Amendments to Liquefied Natural Gas Facilities Advanced Notice of Proposed Rulemaking (ANPRM) published by the Pipeline and Hazardous Materials Safety Administration (PHMSA) on May 5, 2025.[1] Cheniere requests that PHMSA use its authority under 49 USC §60102 and 49 USC §60103 to amend its siting, design, installation, construction, inspection, testing, operation, and maintenance requirements in 49 CFR Part 193 for large-scale liquefied natural gas (LNG) facilities. Our comments focus on three key areas: (1) the mandate under Section 110 of the Protecting our Infrastructure of Pipelines and Enhancing Safety Act of 2020 (PIPES Act)[2] to adopt a risk-based regulatory approach for large-scale LNG facilities, (2) the disadvantages of relying on the prescriptive provisions of National Fire Protection Association (NFPA) 59A[3] and the current 49 CFR 193 text, and (3) the need to differentiate safety regulations between large-scale LNG facilities and other smaller-scale LNG facilities, such as peak shaving facilities.

Beyond the comments included herein, Cheniere supports the comments filed by the American Petroleum Institute, Center for LNG, Interstate Natural Gas Association of America, American Gas Association, American Public Gas Association, and Northeast Gas Association.

I. About Cheniere Energy

Cheniere Energy, Inc. is an international energy company headquartered in Houston, Texas, and is the leading producer of LNG in the United States. Cheniere provides clean, secure, and affordable energy to the world, while responsibly delivering a reliable, competitive, and

[1] Pipeline Safety: Amendments to Liquefied Natural Gas Facilities, 90 Fed. Reg. 18,949 (May 5, 2025).
[2] Protecting our Infrastructure of Pipelines and Enhancing Safety Act of 2020, Pub. L. No. 116-260.
[3] National Fire Protection Association (NFPA) 59A, "Standard for the Production, Storage, and Handling of Liquefied Natural Gas."

integrated source of LNG. Cheniere's operations, construction, and development also support energy and economic development across the United States.

LNG is natural gas in liquid form. It is produced through a refrigeration process that reduces the temperature of natural gas to -260 degrees Fahrenheit, at which point it converts to liquid, enabling global transport in LNG carriers. When used as a fuel, natural gas emits less carbon than coal and oil, with significantly less air pollutants.

Cheniere operates two LNG facilities on the U.S. Gulf Coast. Cheniere's Sabine Pass LNG facility, located in Cameron Parish in southwest Louisiana, currently has six fully operational liquefaction units, or "trains" as they are known in the LNG industry, with an aggregate nominal production capacity in excess of 30 million tonnes per annum (mtpa) of LNG.

Cheniere's Corpus Christi LNG facility, located on the La Quinta Ship Channel, along the north shore of Corpus Christi Bay in San Patricio County, Texas has four fully operational liquefaction trains with an aggregate nominal production capacity, inclusive of authorized expansion projects, in excess of 30 mtpa.

II. Advantages of a Risk-Based Regulatory Approach per the PIPES Act of 2020

Section 110 of the PIPES Act mandates PHMSA to update Part 193 operating and maintenance standards for large-scale LNG facilities (other than peak shaving facilities) to incorporate a risk-based regulatory approach that achieves an equivalent level of safety to current prescriptive standards. We strongly support this directive, as a risk-based approach, as outlined in Section 110 of the PIPES Act, particularly the elements of (c)(1)-(14), offers significant advantages for large-scale LNG facilities. The elements should also be the basis of the implementation plan required under Section 110(d) of the PIPES Act as proposed in ANPRM Section III.G.5.

A risk-based approach – modeled on the elements outlined in Section 110 (c)(1)-(14) and industry standards such as API Recommended Practice 580 (Risk-Based Inspection), API RP 576 (Inspection of Pressure-relieving Devices), API 510 (Pressure Vessel Inspection Code), and API 570 (Piping Inspection Code) – prioritizes safety by focusing resources on higher-risk areas. Data from the Occupational Safety and Health Administration (OSHA) highlights the effectiveness of a risk-based approach. OSHA promulgated a risk-based safety management system at 29 CFR 1910.119 (Process safety management of highly hazardous chemicals) in 1992. Incident Rate data shows that incident rates decreased by 50% from 1992 to 2015. And fatalities and injuries decreased from 4.2/100,000 workers in 1992 to ~2.1/100,000 workers in 2018.[4] This significant incident decrease, demonstrated by a risk-based approach, highlights the effectiveness of a risk-based safety management system. Further demonstrating that a risk-based regulatory approach does not compromise safety, rather it improves safety.

A risk-based approach addresses ANPRM Section III.H as this approach enhances safety outcomes by enabling operators to tailor maintenance and inspection schedules based on facility-specific risk profiles, as opposed to rigid, prescriptive timelines that may not reflect actual

[4] Veriforce. (2024). *Implementing Process Safety Management to Prevent Industrial Disasters.* https://veriforce.com/wp-content/uploads/2024/11/PSM-ASSP-partnership-White-Paper.pdf

operational risks. The maintenance testing frequencies in 49 CFR 193.2619(c) and (d) require prescriptive control system testing timelines, whereas API RP 576 (Inspection of Pressure-Relieving Devices) suggests, if an operator is not utilizing a Risk-Based Inspection (RBI) program, 5- or 10-year intervals for clean, non-corrosive services. Adopting risk-based intervals could reduce costs by millions of dollars annually for a large-scale LNG operator, without compromising safety, by minimizing unnecessary testing that introduces risks through excessive equipment handling. Additionally, control systems testing requires operational downtime. Large-scale LNG facilities operate nearly continuously, and the prescriptive requirements, such as 49 CFR 193.2619(c) and (d) require downtime at six months and sometimes annual frequencies. These shutdowns increase risk, as data indicates higher incident rates during startups and shutdowns.[5,6] By contrast, a risk-based approach minimizes such events, enhancing safety, operational efficiency and reducing environmental impacts, while aligning with Congressional intent under the PIPES Act.

Moreover, prescriptive requirements, such as 49 CFR 193.2635(d) mandate three-year atmospheric corrosion inspections. Aligning with API 570's risk-based intervals (e.g., five years for Class 1 piping) or utilizing an RBI program would reduce the regulatory burden and yield cost savings without any impact to the safety of the facility.

III. Disadvantages of Relying on NFPA 59A and Prescriptive Codes and Standards

We oppose a holistic incorporation of NFPA 59A-2023 into Part 193, as proposed in ANPRM Sections III.D.2, III.D.5–7, due to its prescriptive nature, which conflicts with the PIPES Act's risk-based mandate for large-scale LNG facilities. Large-scale LNG facilities, under a risk-based regulation, such as required by section 110(c)(1)-(14) of the 2020 PIPES Act, would follow those codes and standards, and the editions that are most appropriate for the assets of the LNG facility. LNG facilities should employ accepted engineering practices in developing their processes and procedures. Under a risk-based regulation, facilities would have autonomy to engineer, design, operate, and maintain their facilities according to the best practices at the time as opposed to what will quickly become an outdated set of standards incorporated by reference in 49 CFR 193. Holistic or even partial adoption of later versions of NFPA 59A does not decrease the regulatory burden, as NFPA 59A-2023 includes over 50 prescriptive requirements that duplicate or expand existing 49 CFR 193 provisions, such as Chapter 16's fire protection evaluations (e.g., 16.2.1.3–4) and Chapter 18's prescriptive maintenance mandates (e.g., 18.10.3.1, 18.10.10.6). These examples add regulatory burden without clear safety benefits, potentially increasing costs for large-scale operators without enhancing safety outcomes.

In contrast, a risk-based approach, leveraging standards like API 580, API 576, and the elements outlined in Section 110(c)(1)-(14) of the PIPES Act, allow operators to adopt industry-accepted engineering practices tailored to facility-specific risks. Large-scale LNG facilities already employ elements of Section 110(c) of the PIPES Act, such as maintaining written safety information under Section 110(c)(1), which is provided to PHMSA during the design and

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[5] U.S. Chemical Safety and Hazard Investigation Board. (2018). *Safety Digest: CSB Investigations of Incidents During Startups and Shutdowns.* https://www.csb.gov/assets/1/6/csb_digest_-_startup_shutdown.pdf

[6] Center for Chemical Process Safety (CCPS), Guidelines for Hazard Evaluation Procedures. 2008; p 257.

construction phases of a project and training requirements under Section (c)(7-9) which are reviewed by PHMSA during the operations inspection processes. Additionally, large-scale LNG facilities under Federal Energy Regulatory Commission (FERC) jurisdiction employ processes such as process hazard analyses and management of change. A risk-based approach enables flexibility, fosters innovation, and aligns with the petrochemical industry's proven safety frameworks. PHMSA should develop risk-based regulations for large-scale facilities, avoiding blanket incorporation of NFPA 59A-2023.

IV. Importance of Differentiating Regulations for Large-Scale LNG Export Facilities vs. Smaller-Scale and Peak Shaving Facilities

The ANPRM, particularly Sections III.A.1 and III.B.2, underscores the need to differentiate safety regulations based on LNG facility type due to their distinct risk profiles. Large-scale LNG facilities are nearly continuous operations with larger volumes of storage, extensive geographical footprints (100–1,000+ acres), complex processes, and marine transfer areas for international LNG transport. In contrast, smaller-scale LNG facilities typically support domestic pipelines intermittently and involve simpler processes and smaller footprints.

These material differences – storage capacity, transfer operations, process complexity, and operational continuity – necessitate tailored regulations. NFPA 59A's one-size-fits-all approach fails to account for the diverse risk profiles of LNG facilities. This practice stifles innovation and imposes unnecessary costs, as operators must navigate conflicting standards across agencies. Prescriptive requirements, such as those in NFPA 59A-2023 or current 49 CFR 193, are often overly burdensome for large-scale LNG facilities, which manage complex, continuous operations with robust risk mitigation strategies. For example, large-scale LNG facilities require fewer shutdowns, apart from outages to perform prescriptive maintenance requirements of 49 CFR 193, due to near-continuous operation, reducing risks associated with shutdowns and startups.

Regulatory overlap with agencies like FERC and the U.S. Coast Guard (USCG) presents further complications for large-scale LNG facilities. Clarifying jurisdictional boundaries (e.g., USCG's authority over marine transfer areas per 33 CFR 127.005) and adopting risk-based regulations for large-scale facilities would reduce duplicative oversight and costs, for both the operator and PHMSA. PHMSA should develop regulations that satisfy the PIPES Act Congressional mandate; incorporating risk-based standards aligned with other regulatory and industry frameworks for large-scale LNG facilities.

V. Conclusion

We urge PHMSA to implement a risk-based regulatory framework for large-scale LNG facilities as mandated by the PIPES Act. Holistic adoption of NFPA 59A-2023 for large-scale LNG facilities should be avoided due to its prescriptive nature and misalignment with Congressional intent. Differentiating regulations between large-scale export facilities and small-scale LNG facilities is critical to address unique risk profiles and operational needs. We support PHMSA's efforts to streamline regulations.

Cheniere appreciates the opportunity to submit comments on the record. Should you or your staff have any questions or need any additional information please contact me at robert.smith@cheniere.com. Additionally, representatives of Cheniere are available to meet with you or your staff to discuss our comments in greater detail.

Sincerely,

Rob Smith
Cheniere Energy, Inc.
Vice President, Regulatory Affairs